Korean Petrochemical Industry

Decarbonization Pathway Analysis

Marginal Abatement Cost Curve (MACC) Model

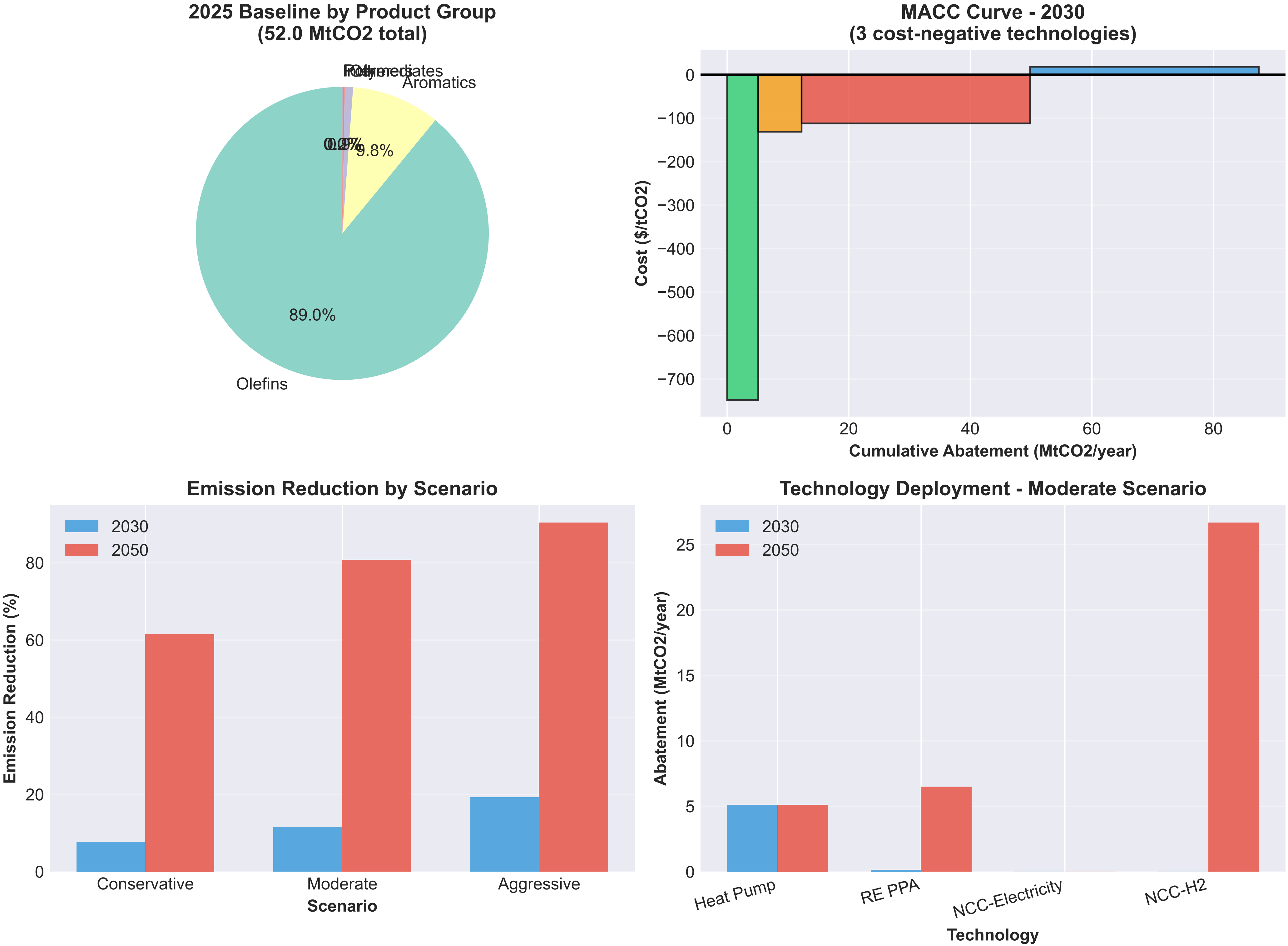
Report Generated: 2025-10-11 23:28

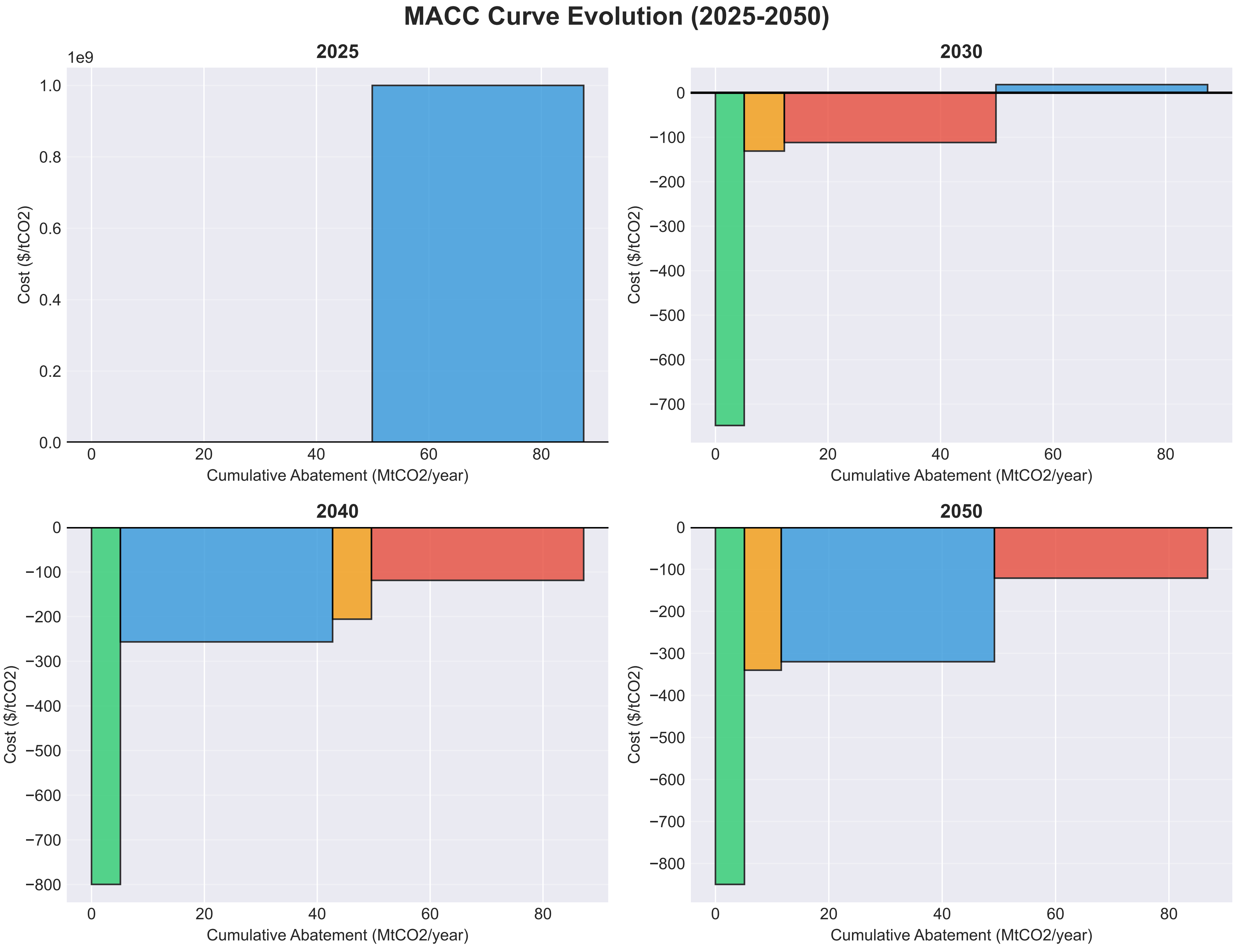
Model Version: 2.2

Key Model Parameters

- Baseline Emissions (2025): 52.0 MtCO2/year
- Facilities Analyzed: 248
- Technologies: 4 (Heat Pump, RE PPA, NCC-H2, NCC-Electricity)
- Time Horizon: 2025-2050
- Maximum Abatement Potential: 49.8 MtCO2 (95.9%)
- Scenarios: Conservative, Moderate, Aggressive

Executive Summary





MACC Curve Evolution (2025-2050)
2025
2030
2040
2050
1e9
Cost ($/tCO2)
Cumulative Abatement (MtCO2/year)
0
20
40
60
80
0.0
0.2
0.4
0.6
0.8
1.0
−100
−200
−300
−400
−500
−600
−700
−800

Emission Trajectories by Scenario

Summary Tables

Scenario Comparison

Scenario	2030 Emis (Mt)	2030 Reduct (%)	2050 Emis (Mt)	2050 Reduct (%)
Conservative	48.0	7.7%	20.0	61.5%
Moderate	46.0	11.5%	10.0	80.8%
Aggressive	42.0	19.2%	5.0	90.4%

Technology Costs - 2030

Technology	Abatement (MtCO2)	CAPEX+OPEX ($/tCO2)	Fuel Diff ($/tCO2)	Total Cost ($/tCO2)
Heat_Pump	5.11	$12.6	$-760.6	$-748.0
NCC-H2	37.60	$24.0	$0.0	$18.3
NCC-Electricity	37.60	$28.7	$0.0	$-111.9
RE_PPA	7.13	$0.0	$-131.1	$-131.1